

Strategies for Estimating Behavioural Frequency in Survey Interviews

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When people answer survey questions of the form “During the past month, how many times did you . . . ?” their responses provide valuable data for researchers and policy makers. Yet the way respondents produce their answers to these “behavioural frequency questions” is not well understood. This article demonstrates that survey respondents can use an array of distinct estimation strategies, depending on what information is available in their memories. The kind of event information that people use is related to factors such as the regularity of occurrence, similarity of one episode to the next, and frequency. In a study conducted as a telephone survey, respondents’ verbal reports and response-time patterns indicate that they usually answer behavioural frequency questions by either retrieving and counting episodes, retrieving or estimating rates of occurrence, or converting a general impression of frequency into a numerical quantity. The third strategy should be of particular concern to survey researchers because respondents provide a quantitative estimate without any relevant numerical knowledge. The set of strategies and the factors that influence their use are integrated into a statistical model that could help survey practitioners to improve data quality and memory researchers to broaden their perspective.

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INTRODUCTION

People estimate various frequencies on an everyday basis. For example, when people seek medical treatment, they are typically asked to describe the frequency of their symptoms. When they consider moving to a new neighbourhood, they might assess how frequently they have heard about crime in that area. When they purchase a new product, one consideration might be how often they believe they will use the product. This article concerns the way people estimate the frequency of autobiographical events when asked about their behaviour on survey questionnaires. We report the results of a study conducted as a telephone survey in which respondents were timed as they answered questions about the frequency of their own behaviour. After answering each question, they described how they arrived at their answer and then rated the regularity and similarity of the target events.

So-called behavioural frequency questions are usually of the form “During the last month how many times did you pay to see a movie?” where both the time frame and event category can vary. The responses to these questions have widespread political and economic implications, so there is social value in understanding how they are formulated. For example, the monthly unemployment rates reported by the US government are based, in part, on responses to behavioural frequency questions on federal surveys. In addition, there are good methodological reasons for studying performance on this task. It is a relatively familiar task for most people and it is carried out in naturalistic settings, with stimuli (autobiographical events) that have personal meaning, yet answering behavioural frequency questions can be studied with some experimental control.

There are relatively distinct bodies of literature on how people answer behavioural frequency questions in survey situations and how they estimate frequency in laboratory tasks. Survey methods researchers (e.g. Blair & Burton, 1987; Burton & Blair, 1991; Means & Loftus, 1991; Menon, 1993) have focused their studies on the use of two response strategies: **episode enumeration (counting all retrieved instances of the event category within the specified time period) and rate-based inferences (recalling that the behaviour happens once a week and reasoning that it must, therefore, have occurred four times in the last month)**. Certain conditions promote the use of one strategy over another and these have been manipulated experimentally (e.g. Blair & Burton, 1987). This is essentially, a dual-strategy view in which responses are based on either counts of retrieved episodes (enumeration) or retrieved rates of occurrence. In this article we present evidence that people use additional strategies for answering these questions but the important point is that survey respondents are assumed to be capable of using multiple strategies.

In contrast, individual laboratory studies have tended to focus on the use of individual strategies. For example, Tversky and Kahneman (1973) demonstrated that people use the ease of retrieving instances as a key to frequency—the

availability heuristic. Hintzman (1988) modelled frequency judgements as a comparison of a target item's similarity to exemplars in memory—a similarity principle. Barsalou and Ross (1986) provided evidence that subjects retrieve and count category instances when reporting category frequency—an enumeration strategy. Although many memory researchers have noted that subjects may estimate frequency in more than one way (Begg, Maxwell, Mitterer, & Harris, 1986; Bruce, Hockley, & Craik, 1991; Hintzman, 1976; Howell, 1973; Johnson, Raye, Wang, & Taylor, 1979; Jonides & Jones, 1992; Jonides & Naveh-Benjamin, 1987; Voss, Verbe, & Bisanz, 1975), the data from laboratory studies point primarily to the use of individual strategies. In a series of laboratory studies that are in many ways analogous to the current survey-format study, Brown (1995) illustrated that individual subjects used multiple strategies to assess the frequency of items that they had recently studied. He used a combination of verbal protocols and response times to differentiate the strategies.

In the current article, we present a study of how respondents answer behavioural frequency questions which is intended to integrate and extend findings from these two bodies of literature. We identify a range of strategies that differ in their conditions of use as well as in their underlying processes, and we suggest that respondents use whichever strategy is best served by the available information.

THE MULTIPLE STRATEGY PERSPECTIVE

Our central idea is that people estimate frequency based on the contents of their memory or, in the absence of relevant content, the state of their memory. In order to count episodes, they must be able to retrieve episodes; in order to infer frequency based on typical rate of occurrence they must be able to retrieve rate information. When they cannot retrieve this type of information, they may be able to use qualitative impressions of frequency, such as “that happened a lot”, converting the impression into a numerical estimate. When none of this information is directly available, people can evaluate the state of their memory to derive frequency information. We term this memory assessment. The availability heuristic (Tversky & Kahneman, 1973) is a good example: The more difficult it is to retrieve instances, the less frequent they are assumed to be. In principle, one can almost always assess the state of one's memory, regardless of whether episodes, rates, or other relevant information can be retrieved. Consequently, we view memory assessment strategies as default procedures.

This view implies that if one could know the value of certain variables, one could at least rule out the use of certain strategies. For example, in order to enumerate episodes, one needs to be able to distinguish one remembered episode from the next. Remembered episodes will be relatively distinguishable if they occur on an irregular schedule and do not have similar characteristics. If one can

determine just how distinctive one episode is from another, it would be possible to determine whether an enumeration strategy is feasible. We discuss certain variables that have been associated with event distinctiveness later.

From a practical viewpoint, this means survey researchers may be able to determine what strategies respondents are capable of using and to rule out others because the necessary information is lacking. This may be useful, for example, in evaluating response accuracy. An enumeration strategy is more likely to lead to underreporting than overreporting because people are more likely to forget individual episodes than to recall episodes that either did not happen or occurred outside the relevant time interval (Sudman & Bradburn, 1974). A methodological intervention that increases reported frequency can be assumed to be increasing response accuracy if one can be sure respondents are enumerating. However, if the episodes about which they are questioned are not distinctive, then enumeration is unlikely and the assumptions about accuracy may need to be questioned. (Note, that we do not address response accuracy in this article but concentrate instead on the processes that underlie frequency estimates.)

Although the multiple strategy view leads to clear predictions about when certain strategies will *not* be used, it is less decisive about when a particular strategy *will* be used. This is because more than one kind of relevant information may be available. For instance, a film enthusiast might know that he or she goes to the movies twice a week (rate information) and might also be able to recall the names and plots of the last several movies attended (episode information); either source of information could be sufficient for an estimate, so the issue is which source respondents will use. Fortunately, there are a number of factors known to affect strategy choice and the way these factors tend to combine makes one strategy more likely to be used than another. Later in the article, we report a statistical analysis in which we examined the combined effect of several variables on the use of particular strategies.

Factors That Affect Strategy Use

Several factors are known to affect strategy use. Blair and Burton (1987) and Burton and Blair (1991) point to the importance of actual frequency in predicting strategy use. They found that episodes tend to be enumerated when frequency is low; that is, when there are relatively few episodes to recall—rate information tends to be used when frequency is higher. Frequency was manipulated by eliciting judgements for either one-month or six-month reference periods, the idea being that frequency is greater for longer time periods. They also found that the time allocated to respond affects strategy selection: more time led to more episode enumeration. Finally, question wording had a weak effect on response strategy: “How often . . . ?” produced

more responses expressed as rates than “How many times...?”. In their discussion, the authors speculate that the regularity and distinctiveness of events may affect strategy as well (Blair & Burton, 1987, p.287; Burton & Blair, 1991, pp.64–65).

The regularity with which events occur has subsequently been shown to affect how respondents produce behavioural frequencies. Means and Loftus (1991) found that activities that presumably occur irregularly, like visiting the doctor for an injury, led to frequency judgements based on recalled incidents more than did activities like visiting the allergist, which presumably occur on a regular schedule. They also showed that similar episodes (minor medical conditions) were recalled less often than distinctive ones (serious medical conditions), although they did not show that such activities led respondents to use rate knowledge instead of enumerating episodes. They did not measure event regularity or distinctiveness, and did not look at both variables together.

Menon (1993) has systematically explored the effects of event regularity and similarity on strategy choice. In a series of studies, she found that events were primarily enumerated when event instances occurred on an irregular basis and when subjects considered them to be distinctive (low in rated similarity). Other levels of these event characteristics led to increased use of rate knowledge. Menon’s perspective differs from our own in that she adopts a strict dual strategy view—respondents either enumerate or use rate information, but there is no role for non-numerical strategies such as those that involve general impressions or memory assessment. However, like Menon (1993), we give considerable weight to the role of regularity and similarity in strategy choice.

Predictions

The multiple strategy perspective leads to several predictions. First, the regularity with which instances of an event occur will be related to the use of rate-based strategies and (indirectly) to the use of enumeration-based strategies. We assume that people are likely to acquire rate information when events occur on a relatively regular schedule. If someone performs a certain activity every Thursday from 2:00 to 3:30, it is likely the person would become aware of this. In contrast, it is unlikely someone would have weekly rate information available for an event that may occur several times in one week, and then not again for several weeks. We predict, therefore, that strategies based on rate information will be used more often for regularly occurring than for irregularly occurring events.

Second, regularity should reduce the accessibility of event instances. This is because a regularly occurring event has identical temporal characteristics each time it takes place, so people may poorly encode the separate episodes. In other words, all else being equal, instances of regularly occurring events should be

more similar to one another than instances of irregularly occurring events. In turn, this implies that instances of regularly occurring events should be more difficult to retrieve than instances of irregularly occurring events, and hence less likely to evoke enumeration. Thus, there are two reasons for predicting that rate-based responses will be more common for regularly occurring events: such events promote the encoding of rate information and the decontextualisation of individual events.

A number of autobiographical memory studies (e.g. Barclay & Wellman, 1986; Brewer, 1988; Linton, 1982; Neisser, 1986; Wagenaar, 1986) have demonstrated that events judged to be similar to many others are hard to recall and recognise. Highly similar episodes lead people to represent common features of those episodes, but not features that distinguish one from the next (Neisser, 1981; Strube, 1987). Therefore, our second prediction is that enumeration should be rare when instances of a target activity closely resemble one another, and should be used more often as the episodes become more distinctive and hence are easier to retrieve.

Our third prediction concerns the times to answer a frequency question. In the current study, response times are used as a means of corroborating the subjects' retrospective strategy reports. We assume that it takes a fixed amount of time to retrieve one episode, so that the more episodes retrieved the longer the response time (Bousfield, Sedgewick, & Cohen, 1954; Brown, 1995). In contrast, we assume that people associate rate knowledge with particular event categories and (as a result) it is directly retrievable. It should take no more time to retrieve a rate of once a day than of once a month, all else being equal. Our third prediction, therefore, is that response times should increase with estimated frequency when respondents enumerate episodes, but there should be no relation between response time and estimated frequency when they use a rate. Similarly, there is no reason, *a priori*, why response times should be related to estimated frequency when respondents use heuristics like availability.

In summary, we are predicting that enumeration will be relatively common when event instances are distinctive and occur at irregular intervals, and that rate-based responses will be prevalent when the events occur at regular intervals. We are also predicting that response times will increase with reported frequency when people enumerate, and that they will not when people rely on rate-based strategies. Note that, although we assume that respondents have multiple strategies available to them, we have focused our predictions on enumeration and rate-based strategies. There is little empirical evidence about what other strategies respondents might use to answer frequency questions on surveys. In light of this, our study has an exploratory character to it as well. We designed the study to identify these additional strategies and to examine which factors (among similarity, regularity, reported frequency, and response time) are related to their use.

METHOD

Design

The study was conducted as a telephone survey. Three experienced interviewers asked 106 respondents to estimate the frequency of 10 autobiographical events and to rate the regularity and similarity of those events.¹ The respondents were contacted from a sample of 250 telephone numbers. These numbers were randomly drawn from a nationally representative sample, based on rural and metropolitan areas of the United States and stratified by Census region and population. If potential respondents declined to participate, interviewers did not attempt to persuade them to take part. Each interview was audiotaped with the respondent's consent.

Procedure

The respondents were presented with four tasks. First they were asked to answer 10 behavioural frequency questions (see Table 1). They were told to think silently before answering, and to take as much time as they needed, but to respond as soon as they had made up their mind. Second, after answering each question, the respondents were asked to explain how they came up with the answer; that is, they were asked for a retrospective protocol (Ericsson & Simon, 1993).² The interviewers occasionally probed if they felt the protocol was not informative.

In the third task, the interviewers presented the 10 events in the Table 1 questions and asked respondents to rate the regularity of each on a 4-point scale. Respondents were told that if they engaged in an event on a routine schedule it was *Very Regular* (4); if they never engaged in it on a routine schedule it was *Very Irregular* (1). The intermediate response options were *Somewhat Regular* (3) and *Somewhat Irregular* (2). The respondents were explicitly instructed not to confuse (in)frequent events with (ir)regular events. The fourth task required respondents to judge the similarity of the events on a 4-point scale. They were told that if their experiences were virtually the same each time they engaged in a behaviour then they should rate it *Very Similar* (4); if each encounter with the event seemed like a totally unique experience, then it should receive a *Very Different* (1) rating. For similarity ratings between these extreme values respondents were given the option of *Somewhat Similar* (3) and *Somewhat*

¹ The event categories were chosen to vary in the similarity and regularity of their individual instances on intuitive grounds; however, this was not systematically controlled or evaluated, and is not discussed further.

² We collected retrospective as opposed to concurrent protocols in order to obtain strategy reports that did not affect the time of the response in the first task. Although retrospective protocols can be a less direct measure of mental process, they do not interfere with the primary task as concurrent think-aloud methods can (Schooler, Ohlsson, & Brooks, 1993).

TABLE 1
Stimulus Questions

During the last month, how many times did you ...

1. conduct a transaction with an Automated Teller Machine?
 2. perform a transaction with a teller in a bank?
 3. shop in a grocery store?
 4. shop in a department store?
 5. purchase gas for your car?
 6. pay to have your car repaired?
 7. receive subscription magazines by mail?
 8. receive catalogues by mail?
 9. eat ice cream?
 10. eat spicy food?
-

Different (2). We settled on the number of points and the particular verbal labels for both the regularity and similarity scales after pilot testing the rating tasks.

This approach to assigning similarity and regularity ratings to stimulus events differs from others in the literature. In particular, Menon (1993) collected ratings from independent judges and established a set of stimuli that were rated either high on both regularity and similarity, low on both, or high on one and low on the other. A particular event was then considered fixed in these characteristics for all respondents. This was effective under Menon's approach in part because her stimulus events were relatively fine-grained (e.g. snacking, washing one's hair) and her subjects were of relatively uniform demographics (undergraduate students). Our stimulus events were somewhat more complicated (e.g. performing a transaction with a teller in a bank), and our respondents more diverse demographically (a stratified, national sample). This means that in our sample, two different respondents might experience events quite differently.

Consider a social security recipient who deposits her cheque at the bank each month when it arrives and interacts with the same teller each time. This event would be quite regular and similar. In contrast, a busy professional whose salary is directly deposited and whose transactions are primarily with an automated teller machine, would be likely to interact with a human teller irregularly (although each interaction may be similar to the next). Under our approach, an event's regularity and similarity were based on within-subject judgements. This meant that one respondent could judge an event to be *Very Regular* and another respondent could rate the same event *Very Irregular*, and subsequent analyses could take into account such individual differences. For this reason, average regularity and similarity ratings are not particularly important in our analyses.

Three different random orders of the events were generated for each respondent, one for eliciting the frequency reports and retrospective protocols, one for the regularity rating task, and one for the similarity rating task. A single

reference period of one month was used for the frequency questions. Our pilot study indicated that respondents found it disruptive to shift reference periods between questions and that zero responses were relative infrequent across the events for this time frame.

Response Coding. The authors listened to a subset of taped interviews and developed a preliminary taxonomy for the response strategies that were evident in the protocols. Two groups of two coders then classified the complete set of protocols, reaching each coding decision by consensus within a group, and expanding the coding taxonomy as needed. Because the groups consistently disagreed on how to code one event (receiving subscription magazines) it was eliminated from further analysis. In a first pass at coding the protocols, the inter-group reliability for the nine remaining items was .75. All discrepancies were then resolved through discussion between the groups so that subsequent analyses were based on complete agreement. The final set of strategies is presented in Table 2.

Response Times. The duration of each taped response was measured from completion of the question by the interviewer to initiation of the spoken frequency by the respondent. Response times included the durations of extraneous speech prior to the actual frequency, for example, qualifiers such as "I'd say about..." and verbalisations of respondents' thinking, such as "Well I do it twice a week, so...". The durations were measured by both groups of coders. As their measurements were highly correlated ($r = .97$, $P < .01$, $n = 950$), only one set of times was further analysed. About 9% of these

TABLE 2
Major Response Categories and Percentage of Observations

<i>Response Strategy</i>	<i>Percentage of Observations</i>
<i>Non-zero Responses</i>	
1. Episode Enumeration	27
2. Rate Retrieval	15
3. Rate Estimation	12
4. Rate and Adjustment	9
5. General Impression	18
6. Uncodable	18
	$n = 660$
<i>Zero Responses</i>	
7. Attempted Enumeration	15
8. Rate Retrieval	80
9. Uncodable	5
	$n = 265$

observations were excluded because respondents asked for clarification or were disrupted before responding.

RESULTS AND DISCUSSION

The overall response rate was 42% which is somewhat low for most surveys. However, the research procedure could have been at least partially responsible: 14% of those contacted were willing to be interviewed but not willing to be tape recorded.³ It is possible that some of those people who were contacted but declined to participate could have been persuaded to take part; however, to reduce interview time and associated cost, the interviewers were instructed not to engage in such “conversion” activities.

Response Strategies

Response strategies were classified separately for frequency reports of zero (26%) and all other values (74%). Looking first at the non-zero responses, 18% of the protocols were uncodable. A total of 96% of the remaining data was assigned to one of five response strategy categories: episode enumeration, rate retrieval, rate estimation, rate and adjustment, general impression. To a certain extent, all but episode enumeration rely on information about the general event category, rather than individual episodes, and so might be reduced to a single rate strategy—and this is what is typically done (e.g. Burton & Blair, 1991; Hubbard, 1992). However, the protocols suggested that the underlying processes were different, so the five distinctions were maintained. The remaining 4% of the codable protocols were assigned to the category of item enumeration (in particular, listing the catalogues typically received in answering the catalogue question). These are not considered further because they were rare and diverged from the phenomenon of interest in that respondents were not concerned with events but rather with items.

Based on reports in the literature, we expected evidence in the protocols of both episode enumeration and of some kind of rate strategies, and both were observed: 27% of the protocols were coded as episode enumeration and 36% were coded as some type of rate-based strategy. An example of an episode enumeration protocol, in response to the question about gas purchases, is “Three. Today and . . . about ten days ago on a job . . . and one other time when I was practically empty.” We subdivided the rate-based responses into rate retrieval (15%), rate estimation (12%), and rate and adjustment (9%). By rate

³ The kind of response-time methodology promoted by Bassili (1993; Bassili & Fletcher, 1991) would be of some help under these circumstances because it times telephone responses “live” as opposed to measuring recorded responses. However, our procedure involved coding retrospective strategy reports which requires a permanent record of the report. For this reason we tape recorded the interviews.

retrieval we are referring to rate information stored some time prior to the interview, e.g. "Four. I go grocery shopping on Fridays." Rate estimation refers to those observations for which respondents compute rate information—as opposed to retrieving it—while answering the question. A rate estimation protocol referred to the respondent's knowledge that the event occurred regularly but with an uncertain frequency, that frequency was somehow constructed in the response process. An example, in response to the automated teller machine question, is "Eight... Last week I went twice so ... in the last month that would be eight."

Respondents were considered to use the rate and adjustment strategy when they relied on rate information (we did not distinguish between retrieved and estimated rates for this strategy) and adjusted up or down to account for exceptions to the rate. An example from the question about grocery shopping is: "Eight... Once a week, regularly, and four more times for ... spontaneous items."

In addition to episode enumeration and the several rate-based strategies, we observed evidence in 18% of the protocols that respondents relied on general impressions. The key characteristic of general impression reports is that respondents use a coarse sense of magnitude to infer a specific frequency. This was usually indicated by so-called vague quantifiers (Wright, Gaskell, & O'Muircheartaigh, 1994)—magnitude terms that have quantitative implications but are not explicitly numerical, for example "all the time", or "a lot". Consider the following report in response to the question about bank teller transactions. "Eight... We're in the process of buying a house *so I've been going there a lot.*" This strategy is fundamentally different from enumeration (and its variants) and rate-based responding (and its variants) in that respondents who lack any numerical frequency information can provide a numerical response by converting their impressions of frequency to a number.

Turning now to the zero responses, 26% of all reported frequencies were zero. Of these, 5% were uncodable. The protocols for the remaining 21% of the responses indicated that respondents use two strategies to determine that they have not engaged in an activity over a given time frame. In particular, the protocols contained evidence of attempted enumeration and rate retrieval. Protocols were coded for attempted enumeration if there was evidence that a memory search failed to locate any episodes within the time period. For example, a protocol provided in response to the ice cream question was, "Zero... Trying to remember if my wife had bought any this month. She hasn't." These were apparent for 15% of zero responses. For 80% of zero responses, respondents indicated that they never engage in the activity or simply knew they had not done so in the previous month. These protocols were coded for rate retrieval. For example, in answering the question about ATM activity, one respondent reported "None ... I don't use them."

Regularity and Similarity Ratings

Consistent with findings reported by Menon (1993), regularity and similarity had clear effects on the choice of response strategy. Mean regularity and similarity ratings for the major response strategies are presented in Table 3. Looking first at non-zero responses, regularity varied with the five response strategies, $F(4,424) = 17.41$, $P < .01$, $MSE = 12.85$.⁴ When respondents enumerated episodes, they judged events to be less regular than when they retrieved rates, $F(1,424) = 54.85$, $P < .01$, $MSE = 40.49$, confirming our first prediction. Moreover, regularity ratings on trials exhibiting episode enumeration were lower than for all other strategies, $F(1,424) = 62.58$, $P < .01$, $MSE = 46.20$, by a Scheffé test. This last effect underscores the idea that events are enumerated primarily when they occur on an irregular basis. Finally, frequency reports relied on general impressions when events were less regular than for the three strategies involving rates, $F(1,424) = 13.54$, $P < .01$, by a Scheffé test. It seems, therefore, that like enumeration, general impressions are used in the absence of rate information. However, as is described in the next section, the frequency conditions under which enumeration and general impressions are used are different.

Similarity showed much the same pattern as regularity. Overall, similarity varied with response strategy, $F(4,423) = 4.39$, $P < .01$, $MSE = 3.04$. In particular, similarity ratings for episode enumeration were lower than for rate retrieval,

TABLE 3
Mean Regularity and Similarity Ratings for the Major Response Strategies

<i>Response Strategy</i>	<i>Regularity*</i>	<i>Similarity**</i>
<i>Non-zero Responses</i>		
1. Episode Enumeration	2.39	2.86
2. Rate Retrieval	3.66	3.54
3. Rate Estimation	3.43	3.31
4. Rate and Adjustment	3.65	3.43
5. General Impression	2.91	3.11
<i>Zero Responses</i>		
6. Attempted Enumeration	1.85	2.46
7. Rate Retrieval	1.40	2.16

* 1 = very irregular, 4 = very regular

** 1 = very different, 4 = very similar

⁴ All ANOVA results reported here are based on models that include a subject term and an item term as classified variables. The subject term is included to account for repeated measures effects and the item term is included to account for item effects. All contrasts reported on the basis of ANOVA are planned unless otherwise noted.

$F(1,423) = 12.54$, $P < .01$, $MSE = 8.70$, confirming the second prediction. Similarity ratings were lower for episode enumeration than for all of the other strategies, $F(1,423) = 16.42$, $P < .01$, $MSE = 11.39$, by a Scheffé test, consistent with the idea that enumeration is used primarily when episodes are quite dissimilar and, consequently, differentiated in episodic memory.

Both event characteristics varied with the strategies observed for zero responses. Events were judged less regular for rate retrieval than for attempted enumeration, $F(1,144) = 5.25$, $P < .05$, $MSE = 2.48$, though both mean ratings are very low (1.40 and 1.85, respectively). Knowing that the event of interest has not occurred in the reference period, respondents may well be mapping their belief that the event could happen in a given month onto the regularity rating scale. By this view, the more plausible it is that an event might occur, the more regularly it is judged to occur. Similarity ratings are low in general, but lower for rate retrieval (2.16) than for attempted enumeration (2.46), $F(1,141) = 4.39$, $P < .05$, $MSE = 3.04$. One explanation is that respondents treat the low end of the similarity scale as a “uniqueness” scale where lower values are more unique: events that are known never to occur would be extremely unique (low similarity) if they were somehow to occur, and events that can plausibly occur would share some characteristics when they do in fact take place, making them less unique (higher similarity).

Regularity and similarity were correlated across the entire data set; that is for both non-zero and zero responses ($r = .48$, $P < .01$, $n = 936$). Although one can imagine scenarios in which regularity and similarity are independent, their correlation is consistent with our earlier contention that events that occur on a regular schedule are experienced much the same from one episode to the next: even if they differ in all other characteristics, each occurrence of a regularly occurring event is temporally similar to the next.

Frequency

Another factor that has been shown to affect which estimation strategies respondents choose is the approximate level of an event's frequency. In particular, subjects are more likely to enumerate low-frequency than high-frequency events in arriving at frequency reports, presumably because the effort of retrieving individual episodes increases with the number of episodes (Blair & Burton, 1987; Burton & Blair, 1991; Means & Loftus, 1991). This is replicated in the current study. Mean frequencies and times for five strategies are presented in Table 4. Overall, frequencies vary with strategy, $F(4,425) = 10.99$, $P < .01$, $MSE = 1153.09$. In addition, frequency for episode enumeration is lower than for rate retrieval, $F(1,425) = 16.27$, $P < .01$, $MSE = 1707.40$ and for all other strategies, $F(1,425) = 41.18$, $P < .01$, $MSE = 4321.43$, by a Scheffé test.

The observation that respondents primarily enumerate low-frequency events (2.3 occurrences per month on average) may well be related to the presence of

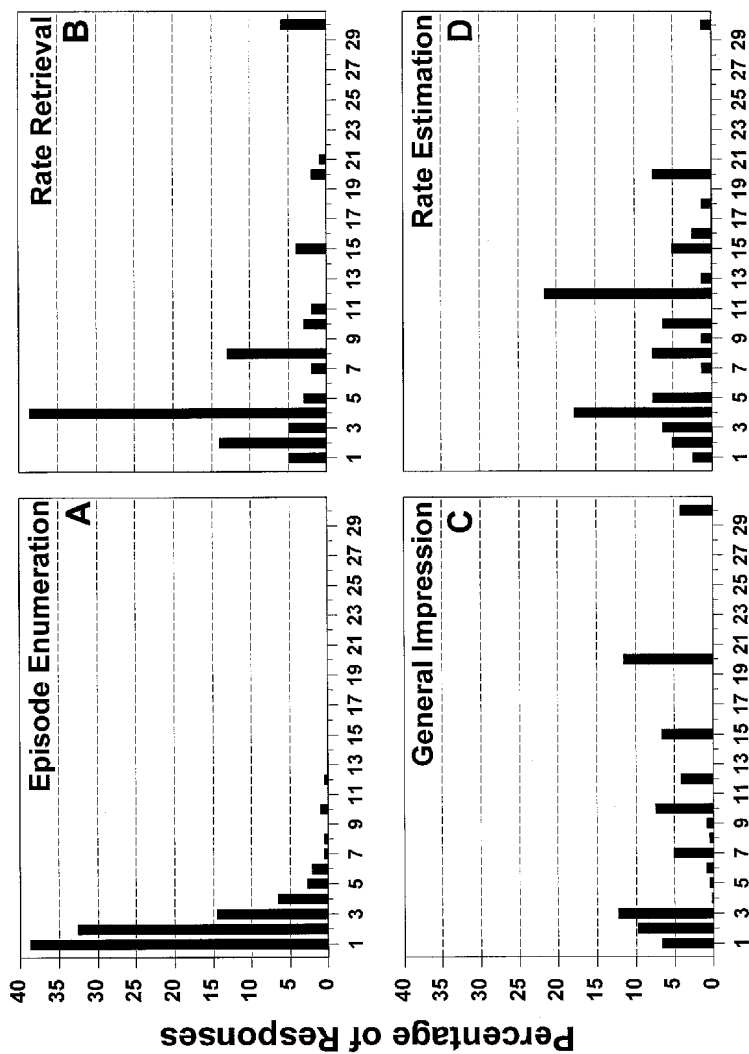
TABLE 4
Mean Frequency and Response Times for the Major
Response Strategies

<i>Response Strategy</i>	<i>Frequency</i>	<i>Time (in seconds)</i>
<i>Non-zero Responses</i>		
1. Episode Enumeration	2.3	4.52
2. Rate Retrieval	7.8	4.36
3. Rate Estimation	11.1	7.28
4. Rate and Adjustment	11.9	6.67
5. General Impression	12.3	6.17
<i>Zero Responses</i>		
6. Attempted Enumeration	0	3.87
7. Rate Retrieval	0	1.58

contextual information for these events. In order to retrieve events, there must be some information to distinguish one event from another. Context can play exactly such a distinguishing role. Because of their repetition, high-frequency events may become decontextualised (Bahrick & Karis, 1982; Linton, 1982; Strube, 1987) and so cannot easily be retrieved.

Respondents used general impressions to report the highest average estimates (12.3 occurrences per month). This is consistent with a finding by Bruce (Bruce & Reed, 1988; Bruce & Van Pelt, 1989) concerning estimates of event frequency on a bicycle trip. The relationship between general impressions and high frequency estimates can be explained if we assume that repeated occurrences allow respondents to directly, albeit imprecisely, encode the fact that particular episodes are frequent. For example, the frequency stored for a repeated event might be "a lot". Infrequently occurring events would be less likely to lead to the impression that the behaviour is rare, because there is simply less opportunity to form this impression. Because respondents do not have precise rate information available when they use a general impression strategy (these events receive the second lowest regularity ratings), all they can do is convert their impression that frequency is high to a relatively large number.

Figure 1 presents the percentages of different frequency responses for episode enumeration, rate retrieval, rate estimation, and general impression strategies. The primary thing to notice is the peaks: when respondents enumerated episodes (Fig. 1a) their estimates were usually small with the number of larger estimates dropping off quickly. The peaks for rate retrieval (Fig. 1b) and rate estimation (Fig. 1d) strategies primarily reflect weekly activity: The peaks at four, eight, and twelve (over the two strategies) suggest that respondents had access to weekly rate information and then multiplied these weekly rates by four (the number of weeks in a month). Although rate retrieval and rate estimation both



Reported Frequency

FIG. 1. Frequency of reported values for four strategies: (a) episode enumeration, (b) rate retrieval, (c) general impression, and (d) rate estimation.

show evidence of converting weekly information to monthly figures, respondents' verbal protocols suggest that the underlying rates have different origins for these two strategies. Retrieved rates are so named because they appear to be stored; estimated rates appear to be the result of partial enumeration. This distinction is further supported by the reaction time data presented in the next section.

When respondents answered on the basis of their general impressions (Fig. 1c) their estimates were often divisible by five. This has the appearance of rounding (Huttenlocher, Hedges, & Bradburn, 1990). It would make sense for respondents using the general impression strategy to produce rounded estimates because they need to convert their impressions, which are qualitative, into numbers; one way to do this is to report a prototypical quantity for the available verbal description, for example, quantifying an impression of "pretty often" as "10". In this domain (as in many others) such prototypical values tend to be divisible by five. One reason that a respondent would map an impression to such a value is that he or she is probably more likely to think of a prototypical than a non-prototypical quantity. Moreover to provide a non-prototypical quantity, say 11, would imply a level of precision that a prototypical value would not necessarily convey; because the conversion from an impression to a number is inherently imprecise, respondents may be more inclined to report the rounded value.

Response Times

Not only do frequency reports vary with response strategies, but they are related to response times in different ways for the various strategies. Response times have been examined in behavioural frequency studies in relation to accuracy (Burton & Blair, 1991) and as an indicator of effort (Menon, 1993). In the current study, response times provide detail about how the different strategies are executed. In particular, we predicted that as frequency reports based on enumeration increase, the latencies for those reports should increase, but frequency and time should not be associated for rate-based reports.

As predicted, the time to report a frequency using this strategy increases with the size of the frequency ($r^2 = .25$, $P < .01$). Also as predicted, there is no relationship between response time and frequency when respondents answered by retrieving a rate. Regression equations, in which response time is a function of estimated rate, are presented for five strategies in Table 5. The differences between the functions for these two strategies indicate that, for enumeration, a noticeable amount of time is associated with the retrieval of each additional episode (about .84 seconds, using the slope of the regression equation for enumeration as an estimate), but for rate retrieval, time does not change with the size of rates; the invariance of time with frequency for rate-based responses suggests that access to rate information requires a fixed number of mental

TABLE 5
Regression Equations for Response Time as a Function of Estimated Frequency

<i>Response Strategy</i>	<i>Equation</i>	<i>r</i> ²	<i>P</i>
Episode Enumeration	time = 2.64 + .84 (reported frequency)	.25	< .0001
Rate Retrieval	time = 4.23 + .02 (reported frequency)	.00	n.s.
General Impression	time = 6.08 + .01 (reported frequency)	.00	n.s.
Rate Estimation	time = 5.56 + .15 (reported frequency)	.14	< .0013
Rate and Adjustment	time = 5.97 + .07 (reported frequency)	.04	n.s.

Time is measured in seconds.

operations—presumably a single retrieval. There was also no relationship between response time and the magnitude of the reported frequencies when respondents used their general impressions. Clearly, general impressions are not based on enumeration. The pattern for this strategy could indicate that impressions are retrieved fully formed. Alternatively, it could suggest that the result of assessing one's memory is a general impression. We cannot distinguish between these interpretations on the basis of the current data.

When respondents based their estimates on rate estimation, their response time increased with the size of the estimates ($r^2 = .14$, $P < .01$) although the relationship does not appear to be as strong as it is for episode enumeration (the slope of the regression equation is .15). Median response times (for non-zero responses) are plotted against reported frequency for episode enumeration and rate estimation in Fig. 2. If there were less than five observations for a particular frequency, that value was omitted from the plot. Both curves show a linear relationship between time and frequency, although the curve for rate estimation is clearly shallower than the curve for enumeration. One explanation for a slope of intermediate steepness is that rates are estimated by enumerating episodes for a portion of the reference period, for example a week; the respondent then generalises to the complete time period much as would be the case with rate retrieval. It is the extrapolation part of the strategy that would weaken the relationship between frequency and time relative to that observed for pure enumeration, because it is possible for a single mental operation (presumably multiplication) to produce a relatively large frequency report.

No relationship between frequency and time is evident for rate and adjustment. Therefore, rate and adjustment may resemble rate retrieval more than rate estimation (where there was a positive relationship between time and frequency). In both rate and adjustment and rate retrieval, the information on which respondents base their estimates is known to them *a priori* and is not estimated through partial enumeration. However, because responses based on rate and adjustment were generally slower than those based on rate retrieval, adjusting the rate seems to require extra time. This suggestion is strengthened by

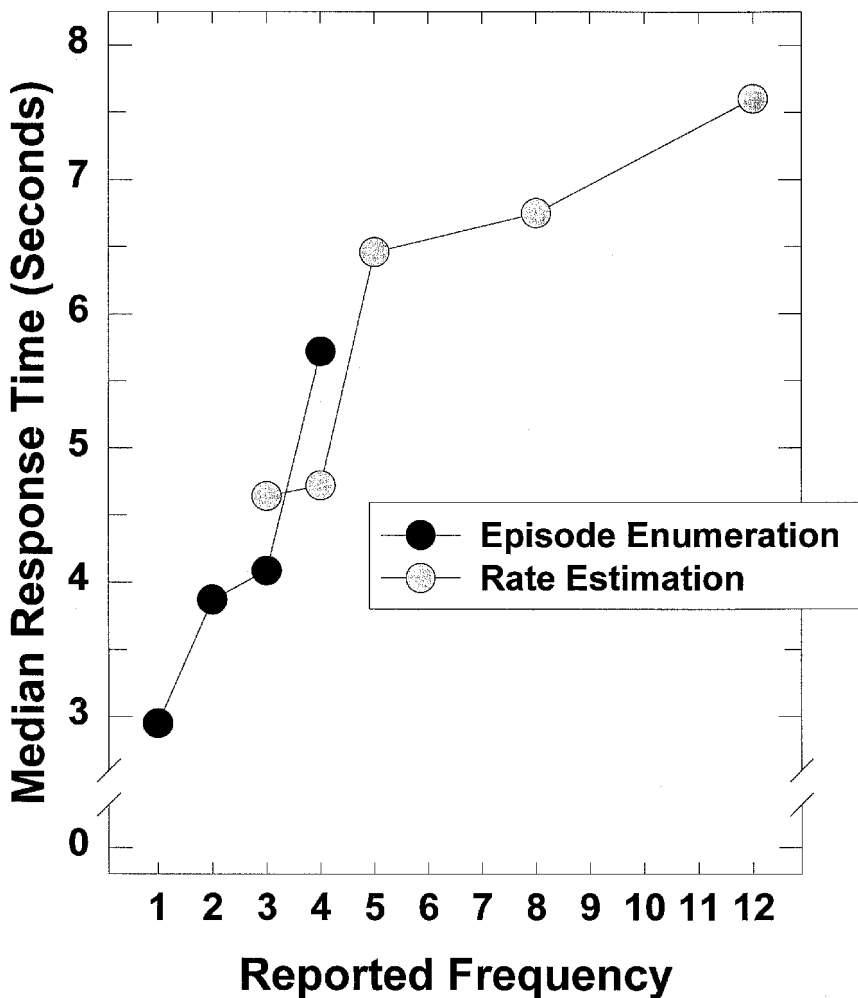


FIG. 2. Median response time as a function of reported frequency and response strategy.

the observation that regularity and similarity judgements for the two strategies are about the same, indicating that they are used under comparable circumstances.

A homogeneity of slopes test confirms that the relationship between time and frequency varies with response strategy, $F(4,368)=7.74$, $P<.01$, $MSE=70.53$. In particular, the regression line for episode enumeration was steeper than that for rate retrieval and general impression, $F(1,368)=20.56$, $P<.01$, $MSE=187.41$. The line for rate estimation was also steeper than that for rate

retrieval, $F(1,368) = 6.20$, $P < .05$, $MSE = 56.47$, by a Scheffé test, but not as steep as the episode enumeration line, $F(1,368) = 14.05$, $P < .01$, $MSE = 128.04$, by a Scheffé test, supporting the idea that the rate estimation strategy is indeed a hybrid of episode enumeration and rate retrieval strategies. These results suggest that respondents may set a threshold for time spent enumerating and once this is exceeded, they extrapolate.

Regardless of the reported frequency, rate retrieval is faster than episode enumeration, $F(1,368) = 12.35$, $P < .01$. This essentially replicates a pattern found by Menon (1993) in which frequency responses for regular items were faster than those for irregular items. And it is also consistent with the finding by Burton and Blair (1991) that inducing respondents to use more time in responding increases the incidence of enumeration. But a lesson of the current study is that overall time tells only part of the story. In arriving at a strategy, respondents are probably influenced by the number of discrete, time-consuming steps they will need to execute in order to respond.

Finally, the two strategies observed for zero frequencies have different response times. Rate retrieval responses were faster than attempted enumeration, $F(1,103) = 30.77$, $P < .01$, $MSE = 78.35$. A simple model for attempted enumeration is that respondents search for a reportable episode for a reasonable amount of time and abandon the search (report “zero”) after that. Based on the regression equation for (non-zero) episode enumeration, time to enumerate one episode is about 3.5 seconds—in the same ballpark as the mean response time of 3.9 seconds for attempted enumeration. One reading is that subjects’ threshold is the time required to enumerate one item. Rate retrieval responses were particularly fast for zero reports; faster, in fact, than non-zero reports based on the same strategy. This difference could arise if non-zero reports demand extra time when a rate expressed in one temporal unit, say weeks, is applied to a reference period expressed in other units, say a month; in contrast, no such extrapolation is required in order to know that an event never occurs and, therefore, report zero.

A Multivariate Approach to Strategy Choice

Each of the variables we have examined in the previous sections (regularity, similarity, reported frequency, and response time) is related to respondents’ choice of estimation strategy. One problem with looking at these relationships individually is that it is hard to see how they jointly lead a respondent to use a particular strategy on a particular occasion. For example, low regularity leads respondents to enumerate when frequency is low; however, low regularity leads respondents to use their general impressions when frequency is high. This relatively simple example involves two variables and two strategies. The patterns become harder to describe when they involve three or four variables and as many strategies. In order to see how particular patterns of several variables

are associated with the use of multiple strategies we performed a discriminant analysis.

Our primary purpose was to describe the combinations of variables that are related to respondents' use of different strategies, not to predict how individual respondents would answer particular questions. In fact, two of the variables in the analysis (response time and reported frequency) were measured as part of the response process and could not be used for prediction purposes. The idea was to form equations from variables such as regularity, similarity, reported frequency, and response times, and use these equations to tell us what strategy was most likely to have been involved in answering each question. It was then possible to evaluate the agreement between the strategy classifications of the discriminant analysis and those of our coders. Good agreement would suggest that the discriminant equations credibly describe the relationships between these variables and strategy choice. Recall that the coders judged what strategy was used to answer a question solely from the content of the protocol, whereas the equations classified an observation solely on the basis of the discriminating variables, blind to the content of the protocol.

In the interest of simplicity, the analysis was limited to three strategies: enumeration, rate retrieval, and general impression. These three appear to involve the most distinct underlying operations, whereas rate estimation and rate and adjust strategies are basically variants of enumeration and/or rate retrieval. The discriminant functions were developed under the assumption that any of the three strategies could occur with equal probability.

The functions were created to differentiate between these strategies on the basis of five discriminating variables: regularity, similarity, frequency, response time, and an indicator variable that was equal to 1 if the estimate was divisible by five and 0 if it was not. The last variable was created because respondents appeared to round their estimates to multiples of five when they relied on general impressions.

Two discriminant functions emerged from the analysis. They were then used to classify the strategy for each observation. The results agreed with the coders' judgements on 65% of the observations. As we have already noted, regularity and similarity are correlated. We might, therefore, be able to remove one of these variables without sacrificing classification power. Similarly, the indicator variable that is concerned with divisibility by five is based on frequency and so those two variables are not independent. Therefore, the functions might be further simplified by removing one of these. To construct more parsimonious functions, we performed a step-wise discriminant analysis. This resulted in two discriminant functions that included regularity, response time and divisibility by five:

$$\text{strategy} = .22 \text{ regularity} + .27 \text{ response time} + .87 \text{ divisible by five}$$

$$\text{strategy} = .98 \text{ regularity} - .31 \text{ response time} - .33 \text{ divisible by five.}$$

The first function is clearly driven by divisibility by five with moderate contributions from the other discriminating variables. It should, therefore, discriminate the use of general impressions from the other strategies because of the rounding that is typical when using this kind of non-numerical information. The second function is dominated by regularity with moderate contributions from the other two terms. This function should be effective at discriminating between rate retrieval and episode enumeration because they are on opposite ends of the regularity range (rate retrieval is highest, episode enumeration is lowest). The moderate response time effect in both functions should help to distinguish between the use of general impressions on the one hand, and episode enumeration and rate retrieval on the other. The average response time for general impressions is 6.17 seconds and the average times for episode enumeration and rate retrieval are 4.52 and 4.56 seconds respectively.

These reduced functions were almost as effective at classifying strategy for each response as were the more complicated functions: the agreement between the functions and coders was 64.4%. The agreement for each of the strategies is presented in Table 6. We interpret this as a reasonable level of agreement because the functions involve only three discriminating variables and still agree with the coders on about two thirds of the observations. A χ^2 test on Table 6 confirms that this pattern of agreement would not be expected by chance ($\chi^2 = 154.94$, $P < .001$), supporting our impression that these discriminant functions successfully differentiate among these strategies.

The discriminant analysis illustrates that several factors simultaneously affect respondents' strategy use. It seems likely that respondents weigh the available information and settle on one or more strategies to pursue. Consider an event that occurs irregularly and often. Because of the event's irregularity, the respondent is not likely to adopt a rate strategy; however he or she may be able

TABLE 6
Classification Results from Discriminant Analysis

<i>Coded Strategy</i>	<i>Number of Cases</i>	<i>Predicted Strategy</i>		
		<i>Enumeration</i>	<i>Rate Retrieval</i>	<i>General Impressions</i>
Enumeration	162	117 72%	37 23%	8 5%
Rate Retrieval	92	18 20%	57 62%	17 19%
General Impressions	108	32 30%	17 16%	59 55%

Top entries in each cell are number of cases; bottom entries are percentage of cases in row.

to recall and count distinct episodes. Because the event occurs frequently, the respondent is likely to have the impression that it occurs frequently. Assuming it is quicker to convert an impression to a number than to enumerate a large number of episodes, the respondent would base his or her estimate on a general impression.

A TAXONOMY OF STRATEGIES

The multiple strategy view is depicted in terms of the underlying processes for each strategy in Fig. 3.⁵ From our perspective, strategy selection is restricted by the contents of memory, but not dictated by them. A strategy may be selected when the relevant information is available, although in several situations other strategies may be applicable as well. However, a given strategy will definitely not be selected when the requisite information is unavailable. This position implies that respondents must often choose between several competing strategies. It is likely that task demands (e.g. time available for a response, importance of an accurate response), context (e.g. previous strategies), and the ease with which different types of relevant information can be recalled, all influence the strategy selection process.

At the highest level we distinguish between enumeration strategies and all other strategies. This reflects our belief that there is a fundamental difference between using remembered episodes and using generic, event-type information as the basis of a frequency report. Two of the strategies we explored appear under the "Enumeration" heading: episode enumeration and rate estimation.⁶ The first of these is the simple retrieve and count strategy. It yields the lowest frequency reports and is used for the most distinguishable and least regular events.

The second strategy, rate estimation, appears under *Enumeration* in Fig. 3 because the relationship between reaction times and frequency reports implies that respondents retrieve individual episodes from a sample portion of the reference period and then extrapolate to the entire period. The strategy seems to be preferred for events of moderate regularity and similarity, and it produces large estimates. These factors, in combination, may lead respondents to stop enumerating before they have retrieved all episodes. It is difficult to recall episodes that are not highly distinctive and respondents may sense that there is a large number of such episodes to retrieve; the regularity of the events gives respondents licence to abandon their enumeration and extrapolate to the full reference period.

The right side of Fig. 3, *Non-enumeration*, is subdivided into strategies that rely on *Direct Retrieval* processes and those that rely on some type of *Memory*

⁵ This is quite similar to the taxonomy that has emerged from laboratory work (Brown, 1995), although the current taxonomy is an extension in that it includes rate strategies.

⁶ We did not observe an enumerate and adjust strategy here even though one of us (Brown, 1995) has observed such a strategy in a laboratory analogue to this study.

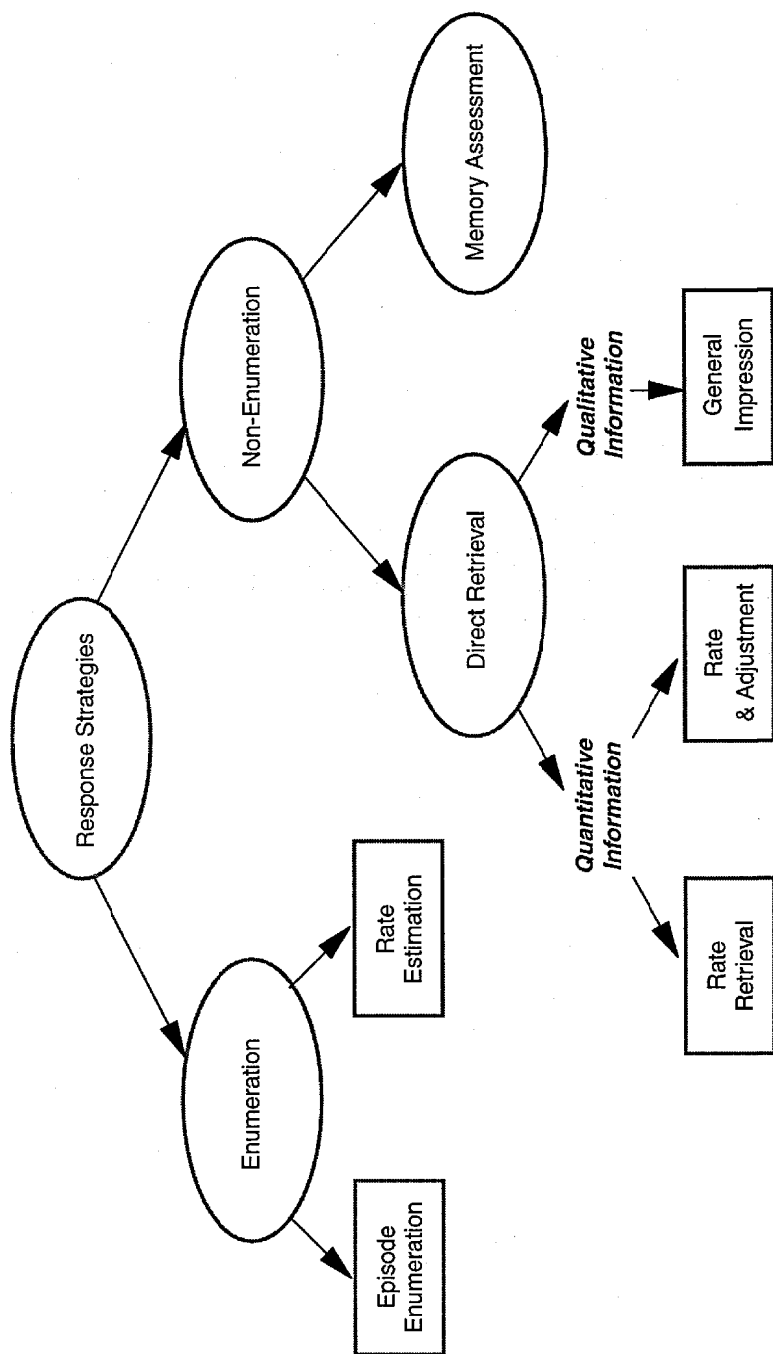


FIG. 3. A taxonomy of strategies for answering behavioural frequency questions.

Assessment. *Direct Retrieval* strategies operate on information that is encoded in respondents' memory before they hear the question.⁷ This stored information can be *Quantitative* or *Qualitative*. The kind of quantitative information that is stored is rate information, for example the knowledge that "I purchase gas several times a week". Both the rate retrieval⁸ and rate and adjust strategies seem to rely on retrieved knowledge of this type, and both are applied to regular and similar events. High regularity is a prerequisite for the availability of rate information and high similarity discourages episode enumeration (Menon, 1993).

In contrast to strategies that use stored quantitative information, a general impression strategy could rely on stored, qualitative characterisations of frequency, such as "I do it all the time". From this perspective, the strategy has two parts: a stored impression must be retrieved and, once it is retrieved, the impression must be converted into an actual number. This conversion process strikes us as a possible locus for the kinds of response scale effects reported by Schwarz and his colleagues (e.g. Menon, Raghbir, & Schwarz, 1995; Schwarz & Bienias, 1990; Schwarz & Hippler, 1987; Schwarz, Hippler, Deutsch, & Strack, 1985). Basically, the finding is that when survey respondents are asked to report frequency judgements using fixed response categories, the size of those categories affects the average estimates. For example, more people report low frequencies of television watching when the scale ranges from "up to $\frac{1}{2}$ hour" to "more than $2\frac{1}{2}$ hours" than when it ranges from "up to $2\frac{1}{2}$ hours" to "more than $4\frac{1}{2}$ hours" (Schwarz & Hippler, 1987; Schwarz et al., 1985). Our view is that without a precise metric to drive the conversion of an impression to a number, respondents may assign extreme impressions like "rarely" and "all the time" to the extreme categories, regardless of the numerical labels for those categories, and estimates they consider to be typical to the middle category, again, without regard to the label.

The far right branch in Fig. 3 is titled *Memory Assessment*. The strategies in this class are distinguished from the others in that they yield frequencies on the basis of judgements about memory processes and states, rather than content, and these judgements are made at the time of responding. In addition to the availability heuristic (Tversky & Kahneman, 1973), other strategies in this class

⁷ This is also true, of course, for all strategies involving enumeration, as individual episodes are directly retrieved.

⁸ Rate retrieval could, in principle, be subdivided into those cases in which the respondent retrieves a rate that is expressed in the same units as the reference period, and those in which a rate is retrieved which is associated with a unit that is smaller than the reference period and is then multiplied to extend to the complete time frame. This is similar to the decomposition strategies (Armstrong, Denniston, & Gordon, 1975) that have been observed by Bradburn, Rips, and Shevell (1987) and Blair and Burton (1987). Because these two variants of rate retrieval rely on the same underlying processes (with the exception of the extrapolation step), and are applied under identical similarity, regularity, and frequency conditions, we have not distinguished between them in this taxonomy.

might take stock of the familiarity of events (Whittlesea, 1993) and the similarity of what is remembered to what is being asked about (Hintzman, 1988; Jones & Heit, 1993).

We have not directly observed memory assessment strategies in our study, and, despite compelling experimental evidence for their use, other researchers in the area report the same curious absence of these strategies (e.g. Blair & Burton, 1987). One reason for this absence might be the limits of verbal protocols—the primary data for identifying strategies in our study and most other studies of behavioural frequency (Blair & Burton, 1987; Burton & Blair, 1991; Means & Loftus, 1991; Menon, 1993). Accurate verbal reports about a mental process require that working memory be involved in the process (Ericsson & Simon, 1993). Because memory assessment does not involve transferring information from long-term memory into working memory—it involves judgements about the state of long-term memory—people cannot verbally report about the strategy (Nisbett & Wilson, 1977; Wilson, 1994). This raises the possibility that some of the uncodable responses (18% of the non-zero responses) were based on memory assessment. In addition, some of the verbal reports that we associate with general impression responses could implicate memory assessment, although we cannot test that here.

CONCLUSIONS

People can answer questions about the frequency of their behaviours using a variety of strategies but the set of strategies that is useful on any one occasion is limited by what information is available. Our multiple strategy perspective makes clear what strategies *cannot* be used when the necessary information is lacking. One practical concern raised by this is the fact that people are able to provide numerical responses when they lack numerical information—that is, when they know virtually nothing about the actual quantities involved. Of the non-zero responses, 18% were based on general impressions. If our proposal about the way people produce these estimates is right, then it does not bode well for the quality of these estimates. Because people can map their impressions to whatever numerical information is available (such as a response scale) or to prototypical values (such as numbers divisible by five), it is unlikely that their estimates are as accurate as survey sponsors assume they are. The logical next step is to study the accuracy ranges for the strategies we have identified in the current study.

We have begun to address the issue of what strategy *is* used when the information required for several strategies is available. Our discriminant analysis provides some of the flavour of plausible decision rules under these circumstances. However, we need an explicit model of how respondents actually choose strategies when they are asked a frequency question. This would provide a more complete theoretical account of the estimation process and could be valuable to survey practitioners.

In conjunction with these additional research activities, our findings can help survey practitioners in two ways. First, it may be possible to develop interventions that help to reduce error for particular strategies. For example, if respondents are likely to enumerate when they answer a particular question then the intervention should be designed to help them recall more episodes, in more detail (e.g. Fisher & Quigley, 1992); if respondents use their general impressions, the intervention might provide helpful numerical references (e.g. Brown, 1995). The particular intervention would be determined by the strategy that is most likely given the patterns of predictive variables.

Second, if it is possible to infer what strategies respondents have used, the survey practitioner can adjust the published frequency statistics following the logic of seasonal (e.g. Bell & Hillmer, 1984) or quality (Armknrecht & Weyback, 1989) adjustment. If one can establish the direction and magnitudes of errors for particular strategies in particular survey domains—potentially by pretesting a small subsample for whom verification data are available—then the data analyst can increment or decrement the reported estimates accordingly.

The goal of bringing memory research to bear on surveys is to improve data quality for the survey practitioner and to produce more realistic theories for the psychologist. The current study is one step in satisfying these goals. How often has interdisciplinary work of this kind actually forged new knowledge? That all depends on how you go about answering the question.

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